Rotation-Invariant Attention Network for Hyperspectral Image Classification

Xiangtao Zheng, *Member, IEEE*, Hao Sun, Xiaoqiang Lu, *Senior Member, IEEE*, and Wei Xie

***Abstract*—*Hyperspectral image* (HSI) classification refers to identifying land-cover categories of pixels based on spectral signatures and spatial information of HSIs. In recent deep learning-based methods, to explore the spatial information of HSIs, the HSI patch is usually cropped from original HSI as the input. And 3 × 3 convolution is utilized as a key component to capture spatial features for HSI classification. However, the 3 × 3 convolution is sensitive to the spatial rotation of inputs, which results in that recent methods perform worse in rotated HSIs. To alleviate this problem, a *rotation-invariant attention network* (RIAN) is proposed for HSI classification. First, a *center spectral attention* (CSpeA) module is designed to avoid the influence of other categories of pixels to suppress redundant spectral bands. Then, a *rectified spatial attention* (RSpaA) module is proposed to replace 3 × 3 convolution for extracting rotation-invariant spectral-spatial features from HSI patches. The CSpeA module, the 1 × 1 convolution and the RSpaA module are utilized to build the proposed RIAN for HSI classification. Experimental results demonstrate that RIAN is invariant to the spatial rotation of HSIs and has superior performance, e.g., achieving an overall accuracy of 86.53% (1.04% improvement) on the Houston database. The codes of this work are available at https://github.com/spectralpublic/RIAN.**

***Index Terms*—Hyperspectral image classification, convolutional neural network, rotation-invariant network, spectral-spatial feature extraction, attention mechanism.**

I. Introduction

HYPERSPECTRAL *images* (HSIs) contain plentiful spectrums and complicated spatial information of the observed scenes [1], [2]. Extensive spectrums of hyperspectral pixels uniquely characterize materials of land-covers in observed scenes, which can be utilized to distinguish various land-covers [3], [4]. HSI classification attempts to recognize the land-cover categories of hyperspectral pixels. However, spectral signatures of hyperspectral pixels are inevitably interfered by atmosphere and instrument noises [5], [6], which pose challenges to HSI classification [7].

Over decades, many methods [8]–[10] have been proposed for HSI classification. Previous works [11], [12] usually leverage hand-crafted feature descriptors (*e.g.* extinction [13] and attribute profiles [14]) to extract features from HSIs. Then, a classic classifier (*e.g.* support vector machine [15]) is employed to obtain land-cover categories of pixels. Recently, various deep networks [16], [17] are proposed for HSI classification. Due to cogent feature representation capabilities, deep networks have become the mainstream research in HSI classification [18].

In HSIs, adjacent pixels may belong to the same land-cover category [19], [20]. Adjacent pixels contain critical spatial clues for HSI classification [9]. Depending on whether adjacent pixels are leveraged, deep network-based methods can be roughly summarized as spectral methods and spectral-spatial methods. The spectral methods [8], [21], [22] usually design 1D deep networks (*e.g. recurrent neural network* (RNN) [23]) to explore spectral signatures of pixels. The spectral methods are invariant to the HSI rotation in the spatial dimension, since these methods only extract features in the spectral dimension and do not involve adjacent pixels [8]. However, spectral methods fail to exploit inherent spatial information of adjacent pixels [24], [25]. The spectral-spatial methods [18], [26] usually study various 2D or 3D convolutional networks to explore the spatial information of adjacent pixels for HSI classification. The spectral-spatial convolution[1] is a key component in recent convolutional networks to capture spectral-spatial features [26], [27]. However, in the channel dimension of the spectral-spatial convolution, all spectral bands of HSIs are regarded equally important for spectral feature extraction. In fact, HSIs usually consist of hundreds of spectral bands, some of which are redundant for HSI classification [27]. Redundant spectral bands of HSIs cannot be effectively eliminated by the spectral-spatial convolution [16].

Manuscript received 21 June 2021; revised 8 March 2022 and 26 April 2022; accepted 12 May 2022. Date of publication 30 May 2022; date of current version 29 June 2022. This work was supported in part by the National Science Fund for Distinguished Young Scholars under Grant 61925112, in part by the Innovation Capability Support Program of Shaanxi under Grant 2020KJXX-091, and in part by the Chinese Association for Artificial Intelligence (CAAI)-Huawei MindSpore Open Fund. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Chaker Larabi. *(Corresponding author: Xiaoqiang Lu.)*

Xiangtao Zheng and Xiaoqiang Lu are with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shaanxi 710119, China (e-mail: luxq666666@gmail.com).

Hao Sun is with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shaanxi 710119, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China.

Wei Xie is with the Hubei Provincial Key Laboratory of Artificial Intelligence and Smart Learning, the National Language Resources Monitoring and Research Center for Network Media, and the School of Computer, Central China Normal University, Wuhan, Hubei 430079, China.

Digital Object Identifier 10.1109/TIP.2022.3177322

[1]The spectral-spatial convolution is 3D form, which consists of 1D channel dimension and 2D spatial dimension. 1D channel dimension is utilized to extract spectral features from spectral signatures, and 2D spatial dimension is used to aggregate spectral features from local neighborhood pixels. If the spatial size of kernels is 1×1, the convolution focuses on the spectral feature extraction and ignores the spatial neighborhood pixels. The convolution with 1×1 kernels is named as the spectral convolution.

1941-0042 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

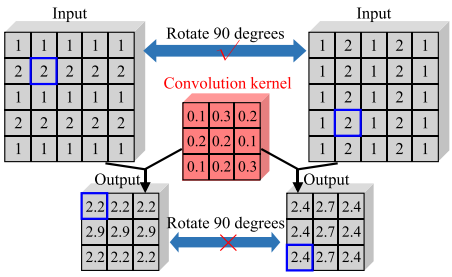

Fig. 1. An example of the spectral-spatial convolution without the padding operation. For convenience, the spectral band number of the input is simply set to 1. Spatial size of convolutional kernel is 3×3. The stride of convolution is 1. "✓" means that one input can be obtained by rotating another input 90 degrees. "×" means that one output does not equal another output after being rotated 90 degrees. Spatial rotation of the input can cause different outputs, which further results in that spectral-spatial methods perform worse in rotated HSIs.

In addition, as show in Fig. 1, the spectral-spatial convolution is sensitive to the spatial rotation of the input image [28], [29]. Due to the overlooking perspective in remote sensing imaging, when the imaging device is rotated, the land-covers in the captured HSIs can been collected at different viewing angle [10], [30]. For the pixel-level HSI classification task itself, rotation should not be a big problem, since rotation does not change the spectral signatures to a large extent. However, in recent spectral-spatial methods, HSI patches containing spatial information are usually utilized as the inputs for HSI classification. The rotation of the imaging device can cause the rotation of the HSI patches, which will cause the performance degradation of recent methods using spectral-spatial convolutions.

In this paper, a *rotation-invariant attention network* (RIAN) is proposed for HSI classification. The proposed RIAN mainly consists of a *center spectral attention* (CSpeA) module and a *rectified spatial attention* (RSpaA) module, which are designed to alleviate the problems of the spectral-spatial convolution in the channel and spatial dimensions, respectively. The proposed CSpeA module is the first layer of the proposed RIAN, which is utilized to suppress redundant spectral bands in a task-driven style. The proposed CSpeA module is leveraged to generate an attention weight to recalibrate all spectral bands, where small values of the attention weight indicate that the corresponding redundant spectral bands are suppressed. The proposed RSpaA module is utilized to extract rotation-invariant spectral-spatial features from recalibrated HSIs. The proposed RSpaA module is based on the similarity among pixels to extract spectral-spatial features, which can avoid the influence of the spatial rotation on the spectral-spatial features.

Similar to previous spectral-spatial methods [8], [10], in the proposed RIAN, adjacent pixels in a square region are cropped from the original HSI to form HSI patches as inputs. In practice, the cropped HSI patches usually contain other land-cover categories of pixels, which can weaken the discrimination of spectral-spatial features [18]. Thus, suppressing the interference from other land-cover categories of pixels is a basic principle of the CSpeA and RSpaA modules. Details of the CSpeA and RSpaA modules are introduced as follows. First, the CSpeA module is proposed to suppress redundant spectral bands of input HSI patches. The proposed CSpeA module is based on the central pixel of the HSI patch to directly generate an attention weight for spectral bands, which can avoid the interference from other land-cover categories of pixels. Second, the RSpaA module explores the similarity among pixels to extract spectral-spatial features. The calculation of the similarity among pixels does not depend on the spatial position of the pixels, making the spectral-spatial features extracted by the RSpaA module invariant to the spatial rotation of HSIs. In addition, a rectified unit is utilized in the RSpaA module to eliminate the interference from other land-cover categories of pixels to the spectral-spatial features. Finally, the CSpeA module, the spectral convolution and the RSpaA module are combined with the residual connection to form RIAN. In experiments, testing samples are rotated to evaluate the performance of RIAN.

The contributions of this paper are as follows:

- Extensive experiments demonstrate that the spatial rotation can weaken the performance of methods utilizing spectral-spatial convolutions as base components.
- A center spectral attention module is proposed to avoid the influence of other categories of pixels to suppress the redundant spectral bands of input HSI patches. The center pixel in the HSI patch is utilized to generate the attention weight for recalibrating spectral bands.
- A rectified spatial attention module is proposed to alleviate the problem of the spatial rotation. The proposed rectified spatial attention module can adaptively capture rotation-invariant spectral-spatial features based on the similarity among pixels.

Rest sections of this paper are as follows. First, Section II reviews recent deep network-based methods. Then, Section III introduces the proposed RIAN. Next, experiments are reported in Section IV. Finally, this paper is concluded in Section V.

II. Related Work

A. Deep Networks for HSI Classification

Based on whether adjacent pixels are exploited for feature extraction, deep network-based methods are roughly summarized as spectral methods and spectral-spatial methods.

Spectral methods [20], [23], [31] make full use spectral signatures to identify the category of hyperspectral pixels. Chen *et al.* [31] employed stacked autoencoder to extract spectral features from pixels. Mou *et al.* [23] took spectral signatures as sequential data and utilized the RNN to analyze sequential data. Hang *et al.* [32] proposed a cascaded RNN to eliminate redundant spectral bands. Wu and Prasad [22], [33] proposed a convolutional RNN to capture contextual features. However, inherent spatial information in HSIs cannot be leveraged by spectral methods [12], [34].

Spectral-spatial methods [35], [36] concentrate on exploring spectral-spatial features from adjacent pixels. Yang *et al.* [37] designed 2D/3D CNNs to leverage spectral correlation and spatial context of HSIs. Lee and Kwon [38] utilized different size kernels to extract contextual information from adjacent pixels. Zhu *et al.* [36] employed a deformable convolution to extract features from adjacent similar pixels. Song *et al.* [39]

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

proposed to fuse hierarchical convolutional features with the residual learning. Deng *et al.* [40] proposed a deep metric model to exploit insufficient labeled pixels for training. Roy *et al.* [41] utilized 2D/3D convolutions to build a hybrid network. Cao *et al.* [26] combined the *convolutional neural network* CNN with a Markov random field to smooth classification results. Zhang *et al.* [1] leveraged six CNN branches to capture contextual interactional features from diverse regions. Hong *et al.* [42] proposed to utilize transformers to model sequence attributes of spectral bands. He *et al.* [43] combined CNN with transformer to explore sequential spectra relationships. In spectral-spatial methods, the spectral-spatial convolution is the most popular operation to extract spectral-spatial features [8], [9], [44]. However, the spectral-spatial convolution is sensitive to the spatial rotation of HSIs [45], [46], which can result in the poor performance of spectral-spatial methods when HSIs are rotated.

B. Attention Mechanism for HSI Classification

Attention mechanism can strengthen effective information (or suppress redundant information) of HSIs for classification [47]. Based on the attention mechanism, recent works [48]–[50] have studied various spectral attention modules and spatial attention modules for HSI classification.

Spectral attention modules aim to suppress redundant spectral bands of input HSI patches or suppress redundant channels of features [16], [27], [51]. Mou and Zhu [16] exploited the global spectral-spatial convolution to generate an attention weight for suppressing redundant spectral bands of HSI patches. Wang *et al.* [51] and Fang *et al.* [27] employed the squeeze-and-excitation module [52] to emphasize informative feature channels.

Spatial attention modules are usually leveraged to facilitate the extraction of spectral features from regions of interest in HSI patches, which can improve the discrimination of features [18], [53]. Sun *et al.* [18] exploited self-attention modules to capture features from homogeneous regions. Wang *et al.* [53] utilized long short-term memory networks to capture key regions for extract attention features. Xue *et al.* [54] exploited attention to generate adaptive weights for multiscale spatial features.

Several works [48], [55] exploit both spectral and spatial attention modules to improve spectral-spatial features of HSIs. Li *et al.* [48] proposed a double-branch network based on the self-attention module, where one branch emphasizes informative feature channels and the other captures spatial contextual information. Yang *et al.* [55] designed cross spectral and spatial attention modules to improve the discrimination of features. Ma *et al.* [56] utilized a multi-layer perceptron to build the spectral attention module and used the spectral-spatial convolution to build the spatial attention module for HSI classification.

Recent attention-based methods have shown excellent performance [16]. However, these attention-based methods still utilize the spectral-spatial convolution as the basic operation, which makes existing attention-based networks still unable to effectively deal with the spatial rotation of HSIs [45].

III. Proposed Method

The proposed RIAN is introduced in this section. First, the overview of RIAN is presented in Section III-A. Then, Section III-B introduces the details of the CSpeA module. Finally, the RSpaA module is described in details in Section III-C.

A. Overview

HSI is the common 3D data, which consists of 2D spatial dimension and 1D spectral dimension [18]. Let $\mathcal{X} \in \mathbb{R}^{W \times H \times C}$ denote the HSI, where $W \times H$ is the spatial size of HSI and C is the band number of spectrums. To obtain land-cover category of the pixel $x_i \in \mathbb{R}^C$, both spectral and spatial information in $\mathcal{X}$ should be considered [8], [57]. In HSIs, due to the spatial smooth prior [7], [26], land-cover categories of adjacent pixels are likely to be the same. Therefore, adjacent pixels of the pixel x_i can be utilized to complement spatial information for classifying the pixel x_i [9]. Adjacent pixels in a square region are cropped out from HSI $\mathcal{X}$ to form an HSI patch $X_i \in \mathbb{R}^{K \times K \times C}$ as the input, where $K \times K$ is the spatial size of the HSI patch. The pixel x_i is located in the center of the HSI patch X_i. Especially, each pixel to be classified has a corresponding HSI patch. Then, RIAN is exploited to infer the category of the pixel x_i based on the HSI patch X_i.

Detailed structures of RIAN are shown in Fig. 2. First, the HSI patch X_i is fed to a CSpeA module to suppress redundant spectral bands of HSI patches. Then, two residual blocks are leveraged to extract the rotation-invariant spectral-spatial features of the HSI patch X_i. Finally, a linear softmax is exploited to obtain the category of the pixel x_i. The residual block of RIAN is composed of two spectral convolutions with *rectified linear unit* (ReLU) and a RSpaA module. The former spectral convolution in the residual block is utilized to captured spectral features from the input of this residual block. The latter spectral convolution in the residual block is utilized to further improve the discrimination of spectral-spatial features from the RSpaA module. The core components of RIAN are the CSpeA module, the spectral convolution and the RSpaA module. The spectral convolution is just the convolution with the spatial size 1×1 kernels, which does not introduce adjacent pixels and can avoid the interference from other categories of pixels in the feature extraction. Detailed configuration of the proposed RIAN is listed in Table I.

B. CSpeA

HSIs usually contain plentiful spectral bands. There will inevitably be redundant spectral bands in the HSI patch X_i [58], [59], which can reduce the classification accuracy. Suppressing redundant spectral bands or enhancing informative spectral bands in the HSI patch X_i can effectively improve classification results [16], [27].

Recent works [16], [27] usually build a spectral attention module to recalibrate spectral bands of the HSI patch X_i. The spectral attention module is embedded into deep networks to generate an attention weight $\alpha \in \mathbb{R}^C$ in a task-driven style [16].

$$\alpha = SpeA(X_i), \tag{1}$$

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

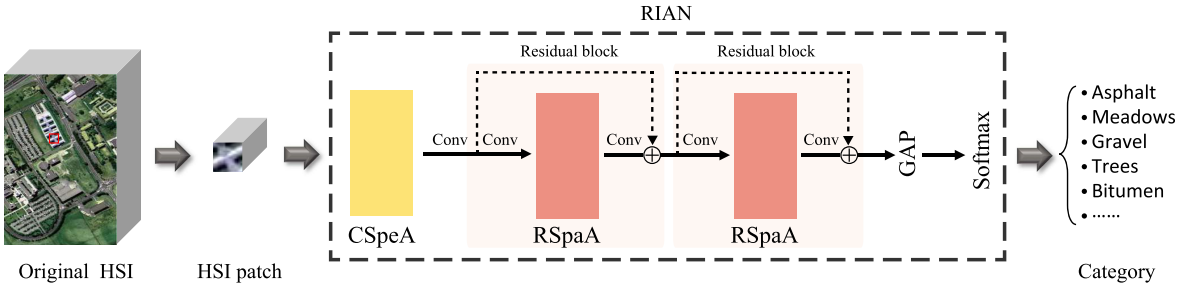

Fig. 2. Framework of the proposed RIAN. The HSI patch is directly cropped from original HSI as the input. First, a CSpeA module is utilized to suppress redundant spectral bands of the HSI patch. Then, two residual blocks are utilized to extract rotation-invariant spectral-spatial features. Finally, a softmax classifier is employed for classification. "Conv" represents the spectral convolution. "$\oplus$" represents the element-wise addition. "GAP" represents the global average pooling.

TABLE I

DETAILED CONFIGURATION OF THE PROPOSED RIAN

Layer	Shape of Input	Shape of Output	Number of Kernels	Connected to
Input	-	(K, K, C)	-	CSpeA
CSpeA	(K, K, C)	(K, K, C)	-	Spectral_Convolution-1
Spectral_Convolution-1	(K, K, C)	$(K, K, 64)$	64	Spectral_Convolution-2, Element-Wise_Addition-1
Spectral_Convolution-2	$(K, K, 64)$	$(K, K, 64)$	64	RSpaA-1
RSpaA-1	$(K, K, 64)$	$(K, K, 64)$	-	Spectral_Convolution-3
Spectral_Convolution-3	$(K, K, 64)$	$(K, K, 64)$	64	Element-Wise_Addition-1
Element-Wise_Addition-1	$(K, K, 64)$	$(K, K, 64)$	-	Spectral_Convolution-4, Element-Wise_Addition-2
Spectral_Convolution-4	$(K, K, 64)$	$(K, K, 64)$	64	RSpaA-2
RSpaA-2	$(K, K, 64)$	$(K, K, 64)$	-	Spectral_Convolution-5
Spectral_Convolution-5	$(K, K, 64)$	$(K, K, 64)$	64	Element-Wise_Addition-2
Element-Wise_Addition-2	$(K, K, 64)$	$(K, K, 64)$	-	Global_Average_Pooling
Global_Average_Pooling	$(K, K, 64)$	$(1, 1, 64)$	-	Fully_Connection
Fully_Connection	$(1, 1, 64)$	(L)	L	Softmax
Softmax (Output)	(L)	(L)	-	-

1. All spectral convolution layers and the fully connection layer are configured with the ReLU and batch normalization.
2. K is set to 9 in experiments. C represents the number of spectral bands in HSI. L represents the number of land-cover categories.

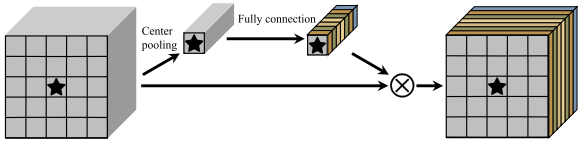

Fig. 3. Illustration of the CSpeA module. $\otimes$ represents the channel-wise multiplication.

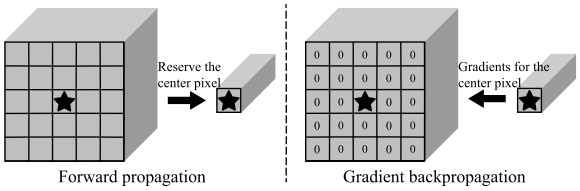

Fig. 4. Illustration of the center pooling. During the forward propagation, the center pooling is utilized to capture the center pixel of inputs. During the backward propagation, gradients of outputs are only propagated back to the center pixel of inputs.

where $SpeA(\cdot)$ represents the spectral attention module. The smaller values in the attention weight α indicate that the corresponding spectral bands are redundant and the larger values indicate that the corresponding spectral bands are informative for HSI classification. Then, the attention weight α is utilized to recalibrate spectral bands of the HSI patch X_i. Recent works [16], [27], [51] usually generate the attention weight α based on the global information of the HSI patch X_i. First, a global spectral-spatial convolution [16] or a global average pooling [27], [51] is utilized to aggregate global information of the HSI patch X_i. Then, a fully connected layer with a sigmoid activation is employed to obtain the attention weight α based on the global information of the HSI patch X_i. However, in practice, the HSI patch X_i often includes interfering pixels, whose categories are different from the center pixel x_i [18]. Interfering pixels in the HSI patch X_i will interfere with the attention weight α, because the informative spectral bands in different categories of pixels are different [16].

In this paper, a simple CSpeA is proposed to avoid the influence of interfering pixels on the attention weight α. The proposed CSpeA generates the attention weight α based on the center pixel x_i of the HSI patch X_i, where the center pixel x_i is the pixel to be classified. Directly using the center pixel x_i to generate the attention weight α can effectively avoid the interference of pixels of other categories. The details of CSpeA are shown in Fig. 3. First, a center pooling is designed to extract the center pixel x_i from the HSI patch X_i. The proposed center pooling is similar to the max pooling. As shown in Fig. 4, during the forward propagation, the center pooling is utilized to capture the center pixel of inputs. During the backward propagation, gradients of outputs are only propagated back to the center pixel of inputs. Then, a fully connected layer with a sigmoid activation is employed to obtain the attention weight α based on the center pixel x_i. Finally, spectral bands of all pixels in this HSI patch are recalibrated by the attention weight α. The proposed CSpeA can be formulated as:

$$x_i = Center(X_i), \tag{2}$$

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

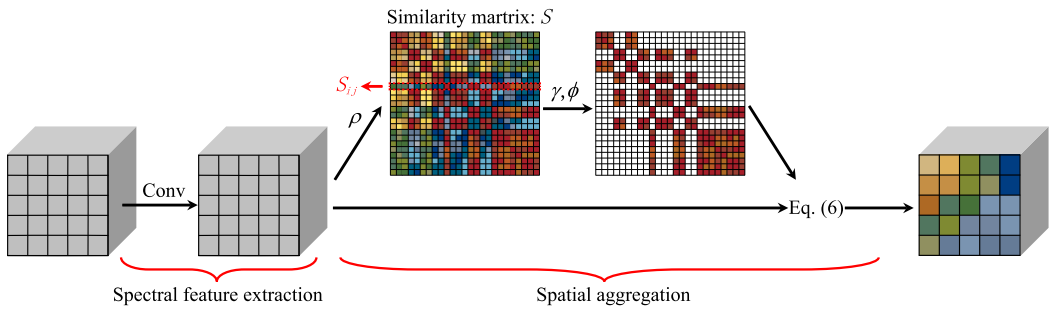

Fig. 5. Illustration of the RSpaA module. Let $S \in \mathbb{R}^{k^2 \times k^2}$ represent the similarities among all pixels. The $((i-1)*k+j)$-th row of the similarity matrix S represents the similarity between all pixels and the (i, j)-th pixel. The $((i-1)*k+j)$-th row of S is obtained by reshaping S_{ij} to $\mathbb{R}^{1\times k^2}$. "Conv" represents the spectral convolution.

$$\alpha = sigmoid(W_c x_i + b_c), \tag{3}$$

$$\hat{X}_i = \alpha \otimes X_i, \tag{4}$$

where $Center(\cdot)$ represents the center pooling, $W_c \in \mathbb{R}^{C\times C}$ is the weight parameter of the fully connected layer, $b_c \in \mathbb{R}^{C\times 1}$ is the bias, $\otimes$ represents the channel-wise multiplication, and $\hat{X}_i \in \mathbb{R}^{K\times K\times C}$ is the recalibrated X_i.

C. RSpaA

In this paper, a RSpaA module is proposed to build a deep rotation-invariant network suitable for HSI classification. The proposed RSpaA mainly focuses on adaptively aggregating adjacent pixel features to generate rotation-invariant spectral-spatial features. Detailed settings of the proposed RSpaA are inspired by the traditional spectral-spatial convolution. To more easily understand the proposed RSpaA, the traditional spectral-spatial convolution is first introduced as follows.

Let $Y \in \mathbb{R}^{K\times K\times C_{in}}$ represent the shallow feature of $\hat{X}_i$, where the spatial size of Y is also $K \times K$, and the channel number of Y is C_{in}. The shallow feature Y is taken as the input to introduce the spectral-spatial convolution. The kernel number of the spectral-spatial convolution is C_{out}. The output of this spectral-spatial convolution is defined as $Z \in \mathbb{R}^{K\times K\times C_{out}}$. The output Z in the position (i, j, c) can be calculated as:

$$Z(i,j,c) = \sum_{m,n\in\Omega_k(i,j), 1\le t\le C_{in}} W_c(i-m, j-n, t) \otimes Y(m,n,t), \tag{5}$$

where $W_c \in \mathbb{R}^{k\times k\times C_{in}}$ represents the c-th kernel, $k \times k$ is the spatial size of kernels, $\Omega_k(i,j) = \{m, n \big| |m-i| \le k/2, |n-j| \le k/2, m \in \mathbb{Z}^1, n \in \mathbb{Z}^1\}$, and the bias is omitted.

As shown in Eq. (5), the spectral-spatial convolution can be divided into the spectral feature extraction ($\otimes$) and the spatial aggregation ($\sum$). In the spectral feature extraction, local $k \times k \times C_{in}$ features are multiplied by $k \times k \times C_{in}$ convolutional kernels in the element-wise style to calculate the spectral feature of each pixel. However, the kernel W_c of the spectral-spatial convolution is not rotation-invariant, which can result in that spectral-spatial convolution is sensitive to the spatial rotation of HSIs. In the spatial aggregation, spectral features of local neighborhood pixels are directly summed to generate spectral-spatial features. However, in the spatial aggregation of the spectral-spatial convolution, spectral features of local neighborhood pixels cannot be adaptively aggregated based on the content of pixels. There may be interfering pixels in the local neighborhood region, which will interfere with the output $Z(i, j, c)$. In addition, limited by the spatial size of kernels, the spectral-spatial convolution can only extract spectral-spatial features from local $k \times k$ neighborhood regions of the HSI patch, and cannot effectively use the global pixels of the HSI patch.

To alleviate the lack of the spectral-spatial convolution, a RSpaA module is proposed to replace the spectral-spatial convolution to build deep networks. Inspired by the spectral-spatial convolution, the proposed RSpaA module is also composed of a spectral feature extraction and a spatial aggregation. Details of the proposed RSpaA module are shown in Fig. 5. In the spectral feature extraction of the RSpaA module, a spectral convolution with ReLU is employed to capture spectral features. Let $\bar{Y} \in \mathbb{R}^{K\times K\times C_{out}}$ represent the spectral features of Y captured by the spectral convolution, where the kernel number of the spectral convolution is C_{out}. In experiments, K is set to 9, and C_{out} is set to 64. Utilizing the spectral convolution in the spectral feature extraction has three advantages. First, the spatial size of kernels is 1×1, which makes the spectral convolution invariant to the spatial rotation of HSIs. Second, compared with the spectral-spatial convolution, the spectral convolution effectively reduces the number of weight parameters. Third, the spectral convolution is easy to implement.

In the spatial aggregation, the proposed RSpaA attempts to adaptively aggregate the spectral features of pixels based on the similarity among pixels. The spectral features of adjacent pixels with high similarity are aggregated to generate the spectral-spatial features, while the spectral features of adjacent pixels with low similarity are not aggregated. The spatial aggregation of the proposed RSpaA can be formulated as:

$$Z(i,j) = \sum_{1\le m,n\le K} \beta_{ij}(\bar{Y}(m,n)) \otimes \bar{Y}(m,n), \tag{6}$$

where $\beta_{ij}(\bar{Y}(m,n)) \in \mathbb{R}^{1\times 1}$ is the attention weight of the (m, n)-th pixel for the (i, j)-th pixel in the HSI patch, $\beta_{ij} \in \mathbb{R}^{K\times K}$ is the attention weight of all pixels for the (i, j)-th pixel, $\bar{Y}(m,n) \in \mathbb{R}^{1\times 1\times C_{out}}$ represents the adjacent pixel, and $Z(i,j) \in \mathbb{R}^{1\times 1\times C_{out}}$. In this paper, $\beta_{ij}(\bar{Y}(m,n))$ is calculated based on the similarity between the (m, n)-th pixel and the

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

(i, j)-th pixel.

$$\beta_{ij}(\bar{Y}(m,n)) = \phi\left(\gamma\left(\rho\left(\bar{Y}(i,j), \bar{Y}(m,n)\right)\right)\right), \tag{7}$$

where $\rho\left(\bar{Y}(i,j), \bar{Y}(m,n)\right) \in \mathbb{R}^{1\times 1}$ is the similarity between $\bar{Y}(i,j)$ and $\bar{Y}(m,n)$, $\gamma(\cdot)$ is a rectified unit that is utilized to make the attention weight of pixels whose similarity is less than the given threshold τ to 0, and $\phi(\cdot)$ is the softmax activation that normalizes the similarity to the attention weight. Accurately measuring the similarity among pixels is important for the spatial aggregation. Recently, many operators have been proposed to measure the similarity between two pixels in convolutional networks, such as: dot product, Euclidean distance and concatenation [60]. These operators have shown similar performance in recent literature [45] and the dot product is the most popular operator [61]. Thus, in this paper, the dot product is employed to measure the similarity between $\bar{Y}(i,j)$ and $\bar{Y}(m,n)$. $\rho\left(\bar{Y}(i,j), \bar{Y}(m,n)\right)$ is calculated as:

$$\rho\left(\bar{Y}(i,j), \bar{Y}(m,n)\right) = \sum \bar{Y}(i,j) \otimes \bar{Y}(m,n). \tag{8}$$

In Eq. (8), a large value of $\rho\left(\bar{Y}(i,j), \bar{Y}(m,n)\right)$ indicates that the similarity between $\bar{Y}(i,j)$ and $\bar{Y}(m,n)$ is high. According to Eq. (8), the similarity $S_{ij} \in \mathbb{R}^{K\times K}$ between all pixels and the (i,j)-th pixel in the HSI patch X_i can be calculated as follows:

$$S_{ij}(m,n) = \rho\left(\bar{Y}(i,j), \bar{Y}(m,n)\right). \tag{9}$$

The goal of the RSpaA is to aggregate spectral features of similar pixels into spectral-spatial features and avoid interference from dissimilar pixels. To drop dissimilar pixels in the spatial aggregation, a rectified unit $\gamma(\cdot)$ is designed to set the similarity of dissimilar pixels to a small value ϵ, which further makes the attention weights of dissimilar adjacent pixels almost 0. The rectified unit $\gamma(\cdot)$ is formulated as follows:

$$\gamma(\mu) = \begin{cases} \mu, & \mu \geq \tau, \\ \epsilon, & \mu < \tau, \end{cases} \tag{10}$$

where τ is the threshold that is set based on the prefixed ratio ν of the maximum value of S_{ij}, and ϵ is set to -100 that makes the attention weight almost 0.

$$\tau = \nu \cdot max(S_{ij}). \tag{11}$$

Finally, the softmax activation $\phi(\cdot)$ is utilized to normalize the rectified similarity to obtain the attention weight β_{ij}, where $\sum \beta_{ij} = 1$.

$$\beta_{ij}(\bar{Y}(m,n)) = \frac{e^{\gamma\left(\rho\left(\bar{Y}(i,j), \bar{Y}(m,n)\right)\right)}}{\sum_{1\leq \hat{m},\hat{n}\leq K} e^{\gamma\left(\rho\left(\bar{Y}(i,j), \bar{Y}(\hat{m},\hat{n})\right)\right)}}. \tag{12}$$

According to Eq. (6) and Eq. (12), the spectral-spatial features can be calculated for each pixel in the HSI patch X_i, respectively. The proposed RSpaA module performs spectral-spatial feature extraction based on the similarity among pixels, making RSpaA invariant to spatial rotation of HSIs.

Both the proposed RSpaA module and the previous self-attention module [18], [48], [60] explore the similarity among pixels to improve features. The differences between the proposed RSpaA module and the previous self-attention module are as follows. 1) The proposed RSpaA module follows the structure of the spectral-spatial convolution, where a spectral convolution is employed to extract spectral features from the input of RSpaA in the spectral feature extraction. However, the self-attention usually employs three different spectral convolutions to generate the query, key and value features for feature extraction, which increases the weight parameters. 2) In the spatial aggregation of RSpaA, a rectified unit is proposed to explicitly avoid the interference from other categories of pixels. However, in the self-attention, the spectral features of all pixels are spatially aggregated, which cannot effectively avoid the interference of other categories of pixels.

TABLE II
NUMBER OF TRAINING/TESTING PIXELS ON THE HOUSTON DATABASE

Class	Name	Training	Testing
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot 1	192	1041
13	Parking Lot 2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
Total		2832	12197

IV. EXPERIMENTS

A. Databases and Evaluation Measures

In this paper, three public hyperspectral databases (Houston,[2] Pavia University[3] and Salinas[3]) are employed to evaluate the proposed RIAN.

Houston database was captured by the ITRES-CASI 1500 sensor in 2012 and was available by the 2013 IEEE GRSS Data Fusion Contest. This database is composed of 349×1905 pixels with 144 spectral bands. The Houston database is interested in 15 categories of land-covers in campus and urban areas. The standard training and testing sample partitions are employed in experiments. Table II reports the number of pixels on the Houston Database.

Pavia University (PU) database was gathered by the ROSIS sensor over Italy in 2002. This database includes 610×340 hyperspectral pixels with 103 spectral bands. This database consists of 9 categories of land-covers in a university. On the PU database, 100 hyperspectral pixels per category are randomly chosen for training. The number of testing pixels is given in Table III.

Salinas database was obtained by the AVIRIS sensor. The Salinas database contains 512×217 hyperspectral pixels with 224 spectral bands. This database is interested in 16 categories of land-covers, most of which are various crops. In each

[2] http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest

[3] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

TABLE III
NUMBER OF TRAINING/TESTING PIXELS ON THE PU DATABASE

Category	Name	Train	Test
1	Asphalt	100	6531
2	Meadows	100	18549
3	Gravel	100	1999
4	Trees	100	2964
5	Metal Sheets	100	1245
6	Bare Soil	100	4929
7	Bitumen	100	1230
8	Bricks	100	3582
9	Shadows	100	847
Total		900	41876

TABLE IV
NUMBER OF TRAINING/TESTING PIXELS ON THE SALINAS DATABASE

Category	Name	Train	Test
1	Brocoli_green_weeds_1	100	1909
2	Brocoli_green_weeds_2	100	3626
3	Fallow	100	1876
4	Fallow_rough_plow	100	1294
5	Fallow_smooth	100	2578
6	Stubble	100	3859
7	Celery	100	3479
8	Grapes_untrained	100	11171
9	Soil_vinyard_develop	100	6103
10	Corn_senesced_green_weeds	100	3178
11	Lettuce_romaine_4wk	100	968
12	Lettuce_romaine_5wk	100	1827
13	Lettuce_romaine_6wk	100	816
14	Lettuce_romaine_7wk	100	970
15	Vinyard_untrained	100	7168
16	Vinyard_vertical_trellis	100	1707
Total		1600	52529

category, 100 hyperspectral pixels are randomly selected to train RIAN and compared methods. Table IV shows the detailed number of pixels.

To measure the performance of RIAN and compared methods, accuracy in each category, *average accuracy* (AA) under different categories, *overall accuracy* (OA) and Kappa coefficient are considered in this paper. Details of these measurements can be found in the literature [18]. Further, visual classification maps are illustrated to qualitatively analyze the performance of different methods.

B. Compared Methods

Since this paper focuses on the generalization performance of deep network-based methods for HSI rotation, several deep network-based methods are employed as compared methods. Details of compared methods are introduced as follows.

- 1-D CNN [21]: It is a spectral method that employs two 1-D convolutional layers to extract spectral features.
- 2-D CNN [62]: Three 2-D convolutional layers are stacked to extract spectral-spatial features.
- DHCNet [36]: The deformable convolution is employed to capture spectral-spatial features from neighboring pixels with similar structures.
- SpecAN [16]: A spectral attention module is proposed to recalibrate the spectral bands of HSI patches. Then, traditional spectral-spatial convolutions are utilized to classify hyperspectral pixels.

TABLE V
OAS (%) OF RIAN WITH DIFFERENT NUMBERS OF CONVOLUTIONAL KERNELS

Number	8	16	32	64	128	256
OA	81.60	84.17	85.32	**85.59**	85.53	85.11

- SSAN [18]: First, several 3-D spectral convolutions are exploited to extract spectral features. Then, the self-attention is employed to generate spectral-spatial features.
- MGCN [63]: A minibatch graph convolutional network is designed to model the spatial relations between hyperspectral pixels with the graph structure for HSI classification.
- DRCNN [1]: First, several CNN branches are utilized to capture contextual interactional features from diverse regions. Then, the fully-connected layers and the softmax classifier are employed for classification.
- CASSN [55]: A cross spatial attention module is designed to extract spatial features under the guidance of the center pixel for HSI classification.

In the data preprocessing, each spectral band of HSIs is normalized to $[0, 1]$ with the min-max scaling. For all methods, the training epoch is 200, and the batch size is 100. Compared methods utilize the learning rate settings reported in original literature. In the proposed RIAN, the initial learning rate is 0.01 and then multiplied by 0.5 after every 20 epochs. 2-D CNN, DHCNet, SpecAN, SSAN, DRCNN, CASSN and the proposed RIAN focus on extracting spectral-spatial features from the HSI patches. The spatial size of HSI patches is set to 9×9 for these methods. 1-D CNN takes hyperspectral pixels as the input to explore the spectral signatures for HSI classification. MGCN takes hyperspectral pixels as the input to explore the spatial relations between hyperspectral pixels with the graph structure for HSI classification.

C. Parameter Settings of RIAN

In this subsection, the Houston database is utilized to acquire the optimal parameter settings for RIAN. And then the same parameter settings of RIAN are used on other databases. The kernel number of spectral convolutions and the number of residual RSpaA blocks are key hyperparameters for the proposed RIAN, which determines the number of weight parameters. The threshold τ determines the number of adjacent pixels that are aggregated to generate spectral-spatial features. The threshold τ is calculated based on the prefixed ratio ν in Eq. (11). For convenience, testing samples without rotation are used to explore the performance of RIAN with different settings.

1) Kernel Number of Spectral Convolutions: In the proposed RIAN, the kernel number of different spectral convolutions is simply set to the same value. The number of rectified residual blocks is set to 3 and the prefixed ratio ν is set to 0.5 to explore the effect of different kernel numbers on the performance of RIAN. The kernel number of spectral convolutions is set from 8 to 256 to conduct experiments, respectively. OAs of RIAN with different kernel numbers are shown in Table V. When the spectral convolution is configured

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

OAs (%) OF RIAN WITH DIFFERENT NUMBERS OF RESIDUAL RSpaA BLOCKS

Number	1	2	3	4	5
OA	85.35	**85.82**	85.59	85.23	85.30

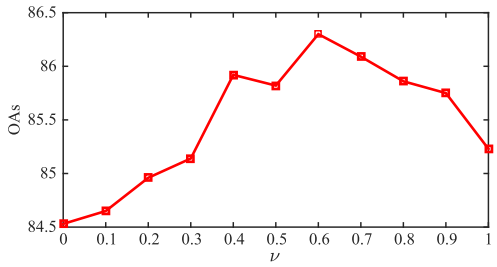

Fig. 6. OAs (%) of RIAN with different prefixed ratio ν.

TABLE VII

OAs (%) OF DIFFERENT ATTENTION MODULES

	Basic	Spectral Attention		Spatial Attention			RIAN
		GSpeA	CSpeA	GC	GSpaA	RSpaA	
OA	75.47	76.86	**79.03**	80.25	80.40	**81.78**	86.30

with 64 kernels, the proposed RIAN achieves the OA of 85.59%.

2) Number of Residual RSpaA Blocks: The effect of different numbers of residual RSpaA blocks on the performance of RIAN is explored in this subsection. The number of residual RSpaA blocks is set from 1 to 5 to conduct experiments, respectively. The kernel number of different spectral convolutions is set to 64 and the prefixed ratio ν is set to 0.5. OAs of RIAN with different number of residual RSpaA blocks are reported in Table VI. As shown in Table VI, RIAN with 2 residual RSpaA blocks achieves the OA of 85.82%. In following experiments, the number of residual RSpaA blocks is set to 2.

3) Threshold τ: In the spatial aggregation of the proposed RSpaA, the threshold τ is also a important parameter. The prefixed ratio ν is manually set to obtain the threshold τ for the spatial aggregation. The value range of the prefixed ratio ν is $(0, 1)$. When the prefixed ratio ν is close to 0, the proposed RSpaA is similar to the traditional self-attention, which aggregates all pixels in HSI patches to generate spectral-spatial features. When the prefixed ratio ν is close to 1, the proposed RSpaA is similar to the spectral convolution, which does not involve adjacent pixels in the feature extraction. The kernel number of different spectral convolutions is set to 64 and the number of rectified residual blocks is set to 2 to conduct experiments. The performance of RIAN with different the prefixed ratio ν is shown in Fig. 6. The prefixed ratio ν is set to 0.6 in following experiments.

4) Effectiveness of the Proposed Attention Modules: To verify the effectiveness of CSpeA and RSpaA, two kinds of experiments on the spectral attention and the spatial attention are carried out, respectively. A simple network composed of a global average pooling and a softmax classifier is utilized as a basic network. The basic network does not introduce any convolution operation and is a linear classification method. To verify the effectiveness of CSpeA, CSpeA is embedded into the basic network to classify HSI. In addition, the *global spectral attention* (GSpeA) module from the literature [16] is employed as the comparison. Different from CSpeA, GSpeA exploits whole pixels of the HSI patch to generate the spectral attention weight [16]. To verify the effectiveness of RSpaA, RSpaA is embedded into the basic network to classify HSI. The *global spatial attention* (GSpaA) module from the literature [18] is also employed as the comparison. Different from RSpaA, GSpaA aggregates whole pixels of the HSI patches to generate spectral-spatial features [18]. Details of experimental networks are as follows.

- Basic network: First, a global average pooling is utilized to fuse whole pixels of HSI patches. Then, a softmax is employed for HSI classification.
- GSpeA network: A GSpeA [16] is added after the input of the basic network to recalibrate spectral bands.
- CSpeA network: A CSpeA is added after the input of the basic network to recalibrate spectral bands.
- *Global convolution* (GC) network: A global spectral-spatial convolution, whose kernel size is the same as the spatial size of input HSI patches, is added after the input of the basic network to extract spectral-spatial features.
- GSpaA network: The query, key and value features of GSpaA [18] are set to the same for fair comparison. The modified GSpaA can be seen as RSpaA without the rectified unit. The modified GSpaA is added after the input of the basic network to extract spectral-spatial features.
- RSpaA network: A RSpaA is added after the input of the basic network to extract rotation-invariant spectral-spatial features.

The spatial size of HSI patches is set to 9×9, kernel number of convolution is set to 64, and the prefixed ratio ν is set to 0.6 to conduct experiments. OAs of different experimental networks on the Houston database are shown in Table VII. The results of GSpeA and CSpeA demonstrate that recalibrating spectral bands can effectively suppress redundant spectral bands and improve the discrimination of HSI patches. The proposed CSpeA can avoid the interference from other categories of pixels and outperforms the previous GSpeA. The results of GC, GSpaA and RSpaA demonstrate that the spatial attention module can effectively extract spectral-spatial features from HSI patches. In the proposed RSpaA, a rectified unit is utilized to avoid interference of other categories of pixels on the spectral-spatial features. The proposed RSpaA performs better than the existing GSpaA.

D. Experiment Settings

Data augmentation with rotations is a common solution to alleviate the problem of HSI rotation [29]. This subsection explores the impact of data augmentation with rotations on the performance of compared methods (1-D CNN, 2-D CNN, DHCNet, SpecAN, SSAN, MGCN, DRCNN and CASSN) on the Houston database. In addition, a *rotation-invariant CNN* (RICNN) [28] is also employed to demonstrate the effectiveness of RIAN. RICNN relies on the data augmentation with rotations for training [28]. In RICNN, a rotation-invariant regularizer is proposed to force a training sample

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
OAs (%) AND TRAINING TIME (SECONDS) OF DIFFERENT METHODS WITH/WITHOUT ROTATION AUGMENTATION ON THE HOUSTON DATABASE

Methods	Without rotation augmentation					With rotation augmentation				
	0	90	180	270	Time	0	90	180	270	Time
1-D CNN	76.63	76.63	76.63	76.63	102.42	77.05	77.05	77.05	77.05	420.95
2-D CNN	77.45	72.08	76.47	71.14	119.51	81.90	81.96	81.71	81.78	487.86
DHCNet	81.53	75.41	80.94	75.04	254.73	85.21	84.86	85.56	85.24	1046.64
SpecAN	84.05	79.47	82.95	79.79	139.07	85.09	84.87	85.18	85.19	567.21
SSAN	82.41	78.07	82.13	79.17	1303.78	85.20	84.92	85.29	85.08	5316.11
MGCN	81.37	81.37	81.37	81.37	102.34	81.57	81.57	81.57	81.57	412.11
RICNN	81.53	75.41	80.94	75.04	254.73	85.53	85.03	85.69	85.36	2142.48
DRCNN	83.83	79.40	83.12	81.05	16474.64	84.65	81.27	84.47	82.57	64752.34
CASSN	85.26	85.26	85.26	85.26	168.06	85.32	85.32	85.32	85.32	673.27
RIAN	86.30	86.30	86.30	86.30	247.47	86.37	86.37	86.37	86.37	992.78

to have the same feature before and after the rotation. The backbone of RICNN is designed for RGB image classification and cannot fully explore the spectral information of HSI for classification. Therefore, we modify the backbone of RICNN as DHCNet [36] to conduct experiments. Firstly, compared methods are trained without/with the data augmentation, respectively. In this paper, data augmentation with rotations refers to that training samples are rotated by 0 (0 means no rotation), 90, 180, and 270 degrees. Then, the performance of compared methods is evaluated in the testing samples rotated by 0, 90, 180 and 270 degrees, respectively.

Experimental results are reported in Table VIII. When the data augmentation with rotations is not employed, the rotation-invariant regularizer in RICNN loses its effect, and the performance of RICNN is equivalent to that of DHCNet. In 2-D CNN, DHCNet, SpecAN, SSAN, RICNN, DRCNN and CASSN, since training samples are HSI patches, data augmentation with rotations can increase the diversity of rotation degrees in training samples. Data augmentation with rotations can improve the performance of 2-D CNN, DHCNet, SpecAN, SSAN, RICNN and DRCNN. In 1-D CNN and MGCN, since training samples are hyperspectral pixels, data augmentation with rotations cannot increase the diversity of hyperspectral pixels. CASSN exploits the cross spatial attention to capture spatial features, which are robust to the rotation. Hence, the data augmentation with rotations cannot significantly increase the performance of 1-D CNN, MGCN and CASSN.

As shown in Table VIII, data augmentation with rotations can alleviate the spatial rotation problem in HSI classification. However, data augmentation with rotations cannot completely solve the problem that 2-D CNN, DHCNet, SpecAN, SSAN, RICNN and DRCNN are sensitive to the spatial rotation, since basic spectral-spatial convolutions are still utilized in these methods. Using rotated samples for training cannot explicitly make spectral-spatial convolutions rotation-invariant. As shown in Table VIII, when the data augmentation with rotations is exploited during the training phase, these methods still have different generalization performance in testing samples at different rotation degrees. In addition, data augmentation with rotations will significantly increase the number of training samples, which leads to the need for more computing resources for training. The RICNN is based on the spectral-spatial convolutions that are inevitably sensitive to the spatial rotation. MGCN cannot fully explore local neighborhood pixels for HSI classification. The proposed RIAN without data augmentation even performs better than compared methods with data augmentation.

In Section IV-E, to fully explore the generalization performance of the proposed RIAN and compared methods to the spatial rotation, the data augmentation with rotations is not employed for all methods. In practice, HSI can be rotated at any degree. However, when rotated to other angles, different HSI patches will contain different adjacent land-covers. When using rotated HSI patches other than 0, 90, 180 and 270 degrees for classification, it is difficult to intuitively analyze whether the changes in the performance of different methods are caused by rotation or different adjacent land-covers. Therefore, testing samples are rotated by 0, 90, 180 and 270 degrees to evaluate the proposed RIAN and compared methods on different databases, respectively.

E. Performance of RIAN and Compared Methods

To evaluate the proposed RIAN, the Houston, PU and Salinas databases are employed. Recent deep network-based methods, including 1-D CNN [21], 2-D CNN [62], DHCNet [36], SpecAN [16], SSAN [18] and MGCN [63], are utilized as compared methods. In this subsection, RICNN [28] is not utilized for comparison, because when data augmentation with rotations is not utilized, the performance of RICNN is equivalent to that of DHCNet.

In addition, two potential networks, the *plain network* (PN) and RIAN\Spe, are further employed as compared methods for ablation analysis. PN represents the proposed RIAN without the CSpeA and RSpaA modules. In PN, the global spectral-spatial convolution is utilized to replace the RSpaA module to extract spatial features. RIAN\Spe represents the proposed RIAN without the CSpeA module. Parameter settings of compared methods are the same as those in the original literature.

1) Houston Database: OAs of 1-D CNN, 2-D CNN, DHCNet, SpecAN, SSAN, MGCN and RIAN at different rotations are shown in Table VIII. The OAs of PN at 0, 90, 180 and 270 rotation degrees are 83.57%, 79.90%, 82.73% and 79.95%, respectively. Since RIAN\Spe is also rotation-invariant, the OAs of RIAN\Spe at 0, 90, 180 and 270 rotation degrees are all 85.02%. When the rotation degree is 0, spectral-spatial methods that can explore spatial information of HSIs show better performance than spectral methods. 1-D CNN and MGCN are invariant to the spatial rotation of HSIs, since 1-D CNN and MGCN directly hyperspectral pixels as the input. At rotation degrees of 90, 180 and 270, the

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

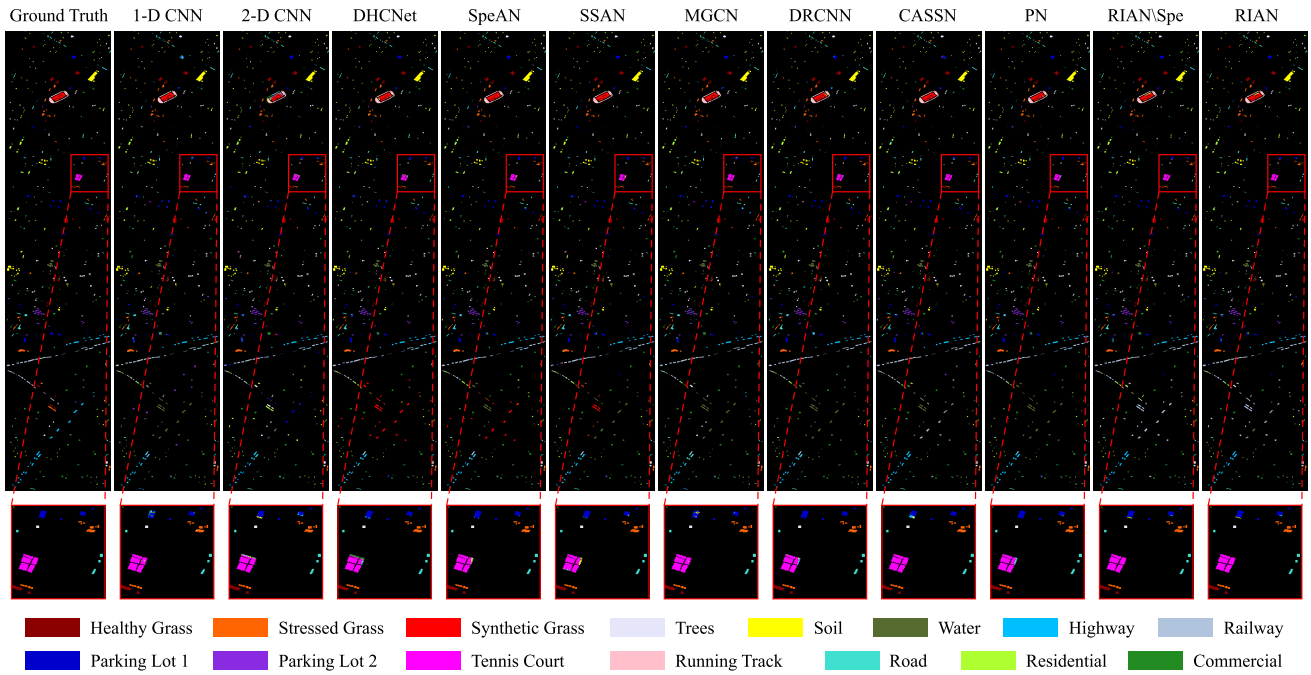

Fig. 7. Classification maps of RIAN and compared methods in testing samples of the Houston database. Best zoomed-in view.

TABLE IX

ACCURACY IN EACH CATEGORY, AA AND KAPPA AT ROTATION DEGREES OF 180 ON THE HOUSTON DATABASE

Class	1-D CNN	2-D CNN	DHCNet	SpecAN	SSAN	MGCN	DRCNN	CASSN	PN	RIAN\Spe	RIAN
1	82.62	81.96	82.05	82.53	82.15	82.43	81.20	**83.10**	82.53	82.81	83.00
2	81.77	82.05	80.92	82.80	83.55	83.27	82.42	**85.15**	82.24	83.74	83.74
3	99.60	63.76	88.71	99.60	98.81	99.41	93.27	**100.00**	94.65	88.32	89.70
4	92.99	96.97	89.11	90.53	92.23	**98.01**	92.05	93.28	91.86	91.19	92.52
5	97.44	96.97	97.35	98.96	98.20	98.86	99.81	99.24	98.96	**100.00**	99.81
6	95.10	97.90	96.50	95.80	95.10	95.10	93.01	95.80	**100.00**	95.80	**100.00**
7	77.33	86.66	85.07	81.81	86.75	76.77	88.62	82.65	82.37	**89.27**	87.59
8	44.06	61.63	71.98	**74.74**	68.00	71.89	73.60	66.48	70.56	72.55	73.22
9	76.68	75.45	**82.53**	80.08	80.93	81.87	81.96	79.60	79.51	81.87	81.21
10	54.63	53.76	54.73	56.47	56.95	48.94	52.70	66.80	60.91	66.60	**68.15**
11	67.93	60.34	64.80	72.30	61.48	76.00	75.33	**90.42**	72.68	87.38	89.85
12	65.03	74.74	85.78	89.72	**93.56**	78.96	87.42	89.15	90.20	80.88	89.15
13	65.96	90.18	89.12	83.51	85.26	57.89	87.72	77.54	83.51	85.61	**92.28**
14	98.38	87.04	74.49	92.71	85.43	99.19	85.43	**100.00**	91.09	**100.00**	**100.00**
15	97.46	55.18	98.73	98.52	94.71	98.73	99.79	**100.00**	93.87	**100.00**	**100.00**
AA	79.80	77.64	82.79	85.34	84.21	83.16	84.95	87.28	84.99	87.07	**88.68**
Kappa	74.84	74.48	79.38	81.58	80.72	79.90	81.80	84.07	81.38	83.73	**85.14**

performance of 2-D CNN, DHCNet, SpecAN, SSAN and PN drops a lot, because these methods exploit traditional spectral-spatial convolution as the basic component. Spectral-spatial convolution makes these methods difficult to deal with the spatial rotation of HSIs. Especially, 2-D CNN and DHCNet perform worse than 1-D CNN at rotation degrees of 90 and 270. SpecAN employs several max pooling layers and fully connected layers to extract features, which result in the loss of the spatial distribution of HSI patches. SSAN employs several 3-D convolutions and self-attention modules to build the network, which introduce too many weight parameters and make training difficult. PN has few weight parameters and achieves comparable results as SpecAN and SSAN. Experimental results of RIAN\Spe show that the proposed RSpaA module can extract rotation-invariant spectral-spatial features. Experimental results of RIAN and RIAN\Spe demonstrate that the proposed CSpeA module can effectively eliminate redundant bands of HSI patches. By suppressing the interference from other land-cover categories of pixels, the proposed RIAN achieves the best OA at different rotations.

To further evaluate RIAN and compared methods, the accuracy in each category, AA and Kappa at the rotation degree of 180 are reported in Table IX. In HSIs, spectral bands contain unique material information of land-covers. Without spatial information of HSIs, 1-D CNN still performs well in the "Healthy Grass", "Stressed Grass", "Synthetic Grass", "Trees", "Soil", "Tennis Court" and "Running Track" categories. With spatial information of HSIs, the classification accuracies of many land-covers have been improved. However, for the "Synthetic Grass" and "Trees" categories, the introduction of spatial information makes several spectral-spatial methods perform worse than 1-D CNN. Adaptive use of spatial information of HSIs to improve classification accuracy still needs further study. In the "Highway" and "Railway" categories, the proposed RIAN performs better than compared methods. Since the proposed CSpeA and RSpaA modules

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

TABLE X

OAs (%) AT DIFFERENT ROTATION DEGREES ON THE PU DATABASE

Rotation	1-D CNN	2-D CNN	DHCNet	SpecAN	SSAN	MGCN	DRCNN	CASSN	PN	RIAN\Spe	RIAN
0	80.95	91.10	96.82	97.77	97.36	86.90	98.33	97.96	96.59	98.31	**98.86**
90	80.95	83.97	90.91	94.97	92.95	86.90	97.09	97.96	93.50	98.31	**98.86**
180	80.95	81.08	90.49	91.39	88.93	86.90	96.70	97.96	89.81	98.31	**98.86**
270	80.95	82.80	90.53	93.56	92.15	86.90	96.41	97.96	91.80	98.31	**98.86**

TABLE XI

ACCURACY IN EACH CATEGORY, AA AND KAPPA AT ROTATION DEGREES OF 180 ON THE PU DATABASE

Class	1-D CNN	2-D CNN	DHCNet	SpecAN	SSAN	MGCN	DRCNN	CASSN	PN	RIAN\Spe	RIAN
1	73.80	72.79	82.22	86.97	87.43	71.43	93.51	97.69	92.05	97.55	**99.04**
2	79.78	88.41	96.22	97.80	96.08	94.79	**98.87**	98.41	92.89	98.51	98.42
3	79.69	68.93	83.84	94.50	91.15	83.29	92.50	96.40	97.65	96.35	**99.35**
4	94.40	92.91	85.29	77.13	66.73	92.07	97.87	**99.39**	86.91	97.10	99.06
5	99.20	98.63	99.92	99.04	99.68	99.20	**100.00**	99.92	**100.00**	99.76	**100.00**
6	74.70	84.50	95.37	96.84	93.81	77.26	99.21	95.11	**100.00**	**100.00**	99.19
7	92.52	76.59	82.68	83.50	88.86	96.26	95.45	99.92	93.98	**100.00**	99.92
8	83.45	47.15	76.38	67.36	58.82	74.62	87.66	97.93	46.45	97.01	**98.91**
9	99.65	75.91	91.38	97.99	99.65	**100.00**	99.88	**100.00**	99.88	99.76	**100.00**
AA	86.35	78.43	88.15	89.01	86.91	87.66	96.11	98.31	89.98	98.45	**99.32**
Kappa	75.36	75.28	87.41	88.51	85.20	82.56	95.60	97.28	86.60	97.75	**98.48**

TABLE XII

OAs (%) AT DIFFERENT ROTATION DEGREES ON THE SALINAS DATABASE

Rotation	1-D CNN	2-D CNN	DHCNet	SpecAN	SSAN	MGCN	DRCNN	CASSN	PN	RIAN\Spe	RIAN
0	89.22	92.03	96.37	95.98	97.21	90.45	96.00	96.79	96.86	97.29	**97.75**
90	89.22	86.70	92.55	91.99	94.94	90.45	94.63	96.79	95.12	97.29	**97.75**
180	89.22	86.14	90.46	90.51	93.27	90.45	94.04	96.79	93.99	97.29	**97.75**
270	89.22	88.29	93.10	92.44	95.15	90.45	95.32	96.79	95.52	97.29	**97.75**

TABLE XIII

ACCURACY IN EACH CATEGORY, AA AND KAPPA AT ROTATION DEGREES OF 180 ON THE SALINAS DATABASE

Class	1-D CNN	2-D CNN	DHCNet	SpecAN	SSAN	MGCN	DRCNN	CASSN	PN	RIAN\Spe	RIAN
1	99.16	99.32	99.63	99.84	99.90	98.95	**100.00**	**100.00**	91.15	**100.00**	**100.00**
2	99.59	98.70	98.15	97.90	99.67	99.83	99.59	99.94	99.83	**100.00**	**100.00**
3	99.57	97.81	94.03	97.33	98.88	99.57	98.88	99.68	98.56	**100.00**	99.89
4	99.61	78.98	95.05	94.90	97.45	99.69	95.67	99.77	**99.85**	**99.85**	99.46
5	97.94	88.05	78.04	90.81	87.20	98.18	95.07	98.80	90.46	**99.61**	99.22
6	99.82	93.91	92.67	91.86	98.03	99.84	99.95	99.97	99.92	**100.00**	99.95
7	99.43	95.66	99.68	98.42	97.24	99.91	98.76	**100.00**	99.89	**100.00**	**100.00**
8	77.01	68.36	84.22	86.51	87.79	79.67	84.44	90.32	83.70	91.10	**93.13**
9	98.95	93.45	98.08	99.77	99.80	99.95	99.39	99.61	**100.00**	**100.00**	**100.00**
10	90.37	83.10	87.92	85.31	89.52	92.89	93.96	98.87	91.44	**99.97**	99.72
11	93.39	73.24	79.75	85.33	84.40	96.90	84.50	99.48	95.76	**100.00**	**100.00**
12	99.89	79.31	92.67	96.72	99.78	99.95	**100.00**	**100.00**	98.08	**100.00**	**100.00**
13	99.39	78.19	68.63	64.83	82.60	99.14	87.25	**100.00**	86.64	**100.00**	**100.00**
14	93.61	96.49	70.21	89.48	88.97	93.30	98.04	99.90	88.76	**100.00**	**100.00**
15	67.05	88.70	89.70	77.93	88.87	68.04	91.55	93.25	**95.77**	94.20	94.80
16	93.67	95.25	98.48	98.77	96.78	98.42	98.89	99.18	99.41	**99.94**	99.88
AA	94.28	88.03	89.18	90.98	93.55	95.27	95.37	98.67	94.95	99.04	**99.13**
Kappa	87.98	84.60	89.37	89.42	92.49	89.35	93.37	96.42	93.30	96.98	**97.49**

facilitate RIAN to extract discriminative features, the proposed RIAN outperforms compared methods in terms of AA and Kappa.

Visual classification maps on the Houston database are shown in Fig. 7. The shapes of the edges in various regions are often irregular. The spatial rotation can change the spatial structures of HSI patches cropped at the region edges. Therefore, the misclassified pixels of previous spectral-spatial methods are mostly located at the region edges. The proposed RIAN is invariant to the spatial rotation and has superior performance at the region edges.

2) PU Database: OAs of RIAN and compared methods at different rotations are shown in Table X. Similar to the Houston database, when the testing samples are rotated, the results of recent spectral-spatial methods decrease significantly. 1-D CNN, MGCN, RIAN\Spe and RIAN are invariant to the spatial rotation. When the rotation degree is 90, 180 and 270, the proposed RIAN is much better than compared methods in terms of OA.

The accuracy in each category, AA and Kappa of RIAN and compared methods at the rotation degree of 180 are reported in Table XI. In DHCNet, SpecAN and SSAN, the introduction of spatial information significantly improves the accuracies of the "Asphalt", "Meadows", "Gravel" and "Bare Soil" categories. However, in the "Trees", "Bitumen", "Self-Blocking Bricks" and "Shadows" categories, 2-D CNN, DHCNet and SpecAN fail to effectively explore the spatial information of rotated HSI patches, and even perform worse than 1-D CNN. The proposed RIAN is invariant to the spatial rotation and performs better than recent spectral-spatial methods.

Visual classification maps are given in Fig. 8. The classification maps of 1-D CNN and MGCN are not smooth, since 1-D

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

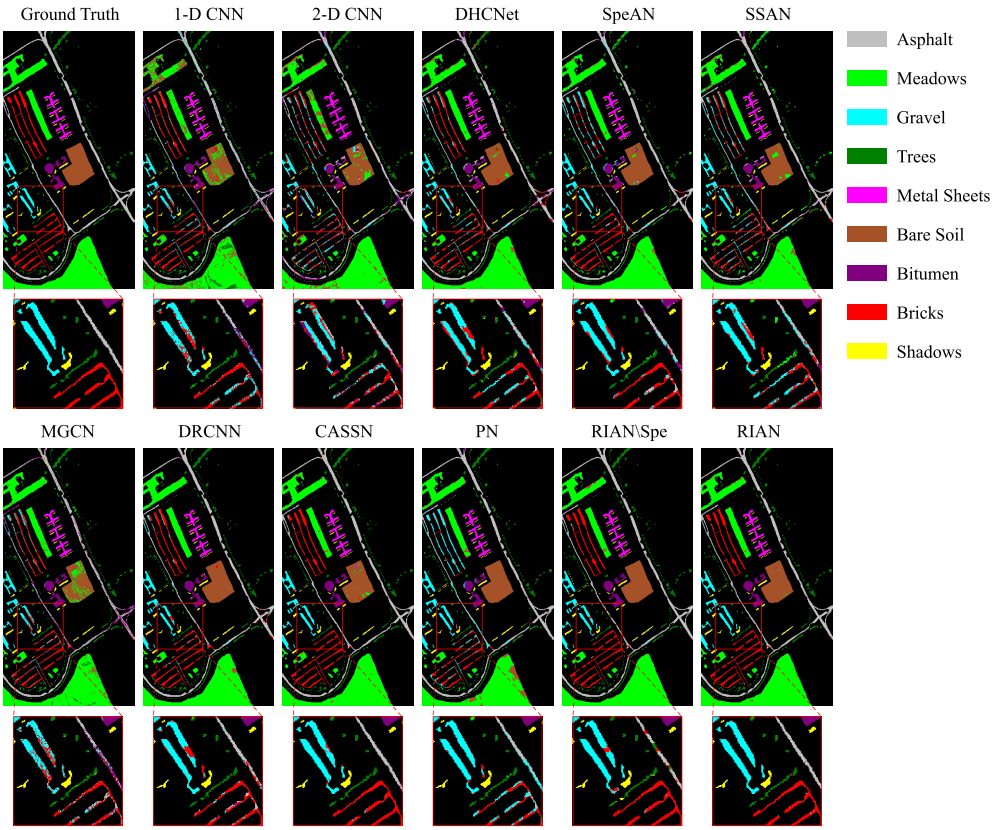

Fig. 8. Classification maps of RIAN and compared methods in testing samples of the PU database. Best zoomed-in view.

CNN and MGCN do no exploit local spatial information of HSI. In other compared methods, local spatial information of HSI is explored for classification. Thus, the classification maps of these methods are smooth. However, these methods perform worse at the edges of regions, especially in the "Meadows" and "Bare Soil" categories, since spectral-spatial convolutions are sensitive to the spatial rotation of inputs. The classification map of RIAN is smooth.

3) Salinas Database: OAs on the Salinas database at different rotations are shown in Table XII. The proposed RIAN outperforms compared methods at different rotation degrees. The accuracy in each category, AA and Kappa at the rotation degree of 180 are listed in Table XIII. In the "Fallow_smooth", "Lettuce_romaine_4wk" and "Lettuce_romaine_6wk" categories on the salinas database, 2-D CNN, DHCNet, SpecAN and SSAN cannot extract effective features from rotated HSI patches. The proposed RIAN can extract discriminative features from rotated HSI patches with the CSpeA and RSpaA modules. The proposed RIAN also outperforms compared methods on the Salinas database in terms of AA and Kappa.

Visual classification maps are shown in Fig. 9. Classification maps of spectral-spatial methods are smoother than that of 1-D CNN. Compared methods show weak performance at the edges of the "Corn_senesced_green_weeds" and "Lettuce_romaine_4wk" categories. By extracting rotation-invariant spectral-spatial features, the classification map of RIAN is smooth and has superior performance at region edges.

4) Discussion: On Houston database, when the testing HSI is not rotated, the proposed RIAN can achieve 86.30% overall accuracy, while recent methods [42], [63] can achieves 88.01% or even 88.62% overall accuracy. However, recent works [42], [63] do not focus on the problem that 3×3 convolutions are sensitive to image rotation and still utilize 3×3 convolutions to build various deep networks. These methods [42], [63] cannot effectively deal with the problem of HSI rotation. In this paper, we propose a rotation-invariant attention network for HSI classification. In fact, the performance of the proposed RIAN can also be competitive with these methods when the image is not rotated. In the future, we will further investigate rotation-invariant deep network structures for HSI classification.

Recent graph neural network-based methods usually extract spatial features through graph structure, which are also robust to image rotation. However, these methods usually use the spectral signatures of hyperspectral pixels as nodes to build the graph structure, which cannot effectively suppress its redundant spectral bands. In addition, graph neural network-based methods usually focus on extracting the spatial features from long-range spatial information of HSI. These methods cannot effectively exploit locally spatial information of HSI, which is important for HSI classification, because adjacent pixels have a high probability of sharing the same land-cover categories. The proposed RIAN shows better performance than the recent graph neural network-based method MGCN.

5) Time Cost: The time cost of RIAN and compared methods is reported in Table XIV. For fair comparison, all methods are reimplemented on a workstation, which is configured with the Ubuntu system, NVIDIA Titan X, Intel Core i7-5930K. SSAN takes many 3-D spectral convolutions

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

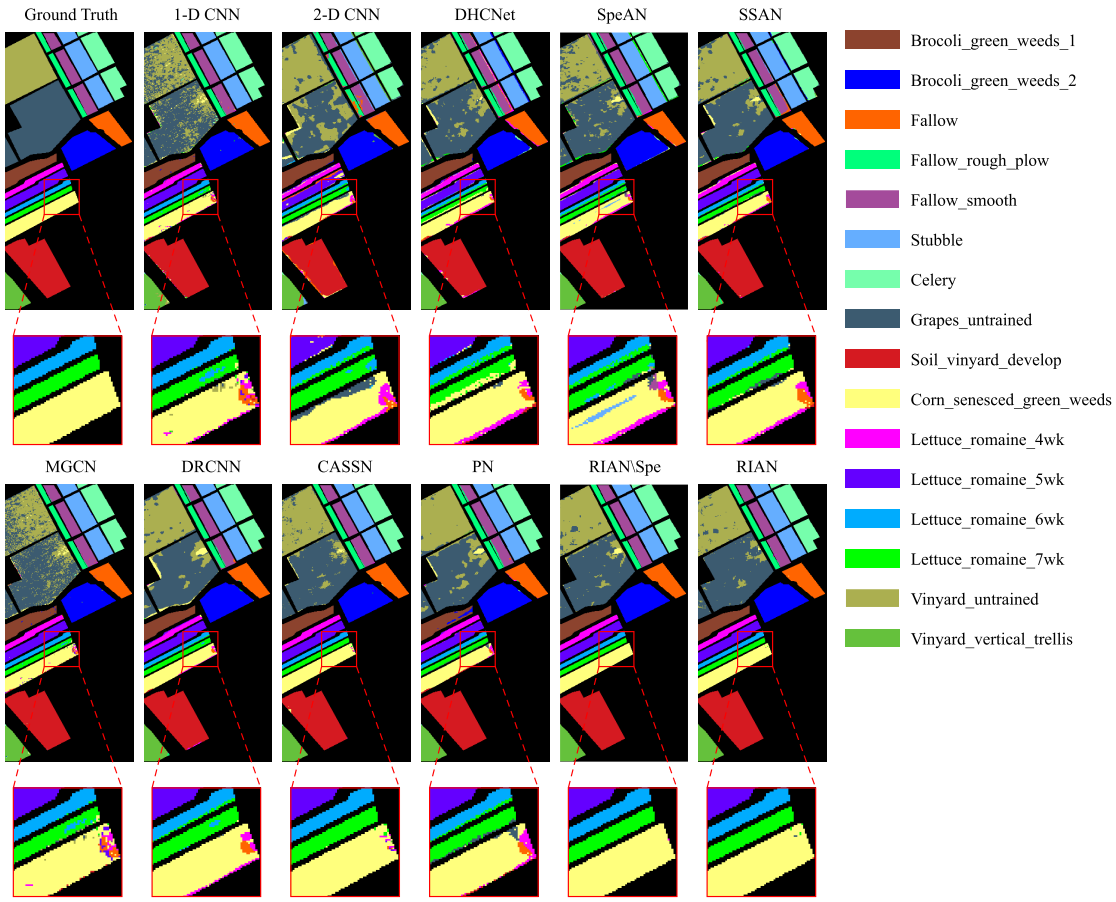

Fig. 9. Classification maps of RIAN and compared methods in testing samples of the Salinas database. Best zoomed-in view.

TABLE XIV
TIME COST (SECONDS) ON EACH HSI DATABASE

Methods	Houston		PU		Salinas	
	Train	Test	Train	Test	Train	Test
1-D CNN	102.42	1.05	26.08	3.16	87.53	7.47
2-D CNN	119.51	1.34	31.15	3.56	98.57	8.46
DHCNet	254.73	2.91	76.88	9.28	175.08	14.32
SpecAN	139.07	1.42	38.13	3.93	110.57	8.80
SSAN	1303.78	8.89	308.43	21.68	1116.68	57.25
MGCN	102.34	1.31	224.14	11.63	391.42	18.08
DRCNN	16474.64	99.35	11628.19	231.71	13315.46	264.57
CASSN	168.06	1.51	54.60	4.35	146.69	9.80
PN	236.91	2.67	71.58	7.89	164.95	13.48
RIAN\Spe	242.48	2.87	72.04	8.03	169.79	14.39
RIAN	247.47	2.95	73.27	8.29	171.34	14.42

as basic operations, which result in the low computational efficiency. DRCNN contains six CNN branches, each of which needs to be trained. This leads to the low computational efficiency of DRCNN. The time cost of SpecAN is close to that of 2-D CNN, because SpecAN has a simple and efficient spectral attention module. RSpaA calculates the graph structure in small-sized HSI patches, and does not greatly increase the computational complexity. The FLOPs of RSpaA are $K^2(2C_{in}C_{out} - C_{out}) + 4K^4(C_{out} + 1)$. Therefore, the proposed RIAN does not significantly increase the training and testing time while maintaining the superior performance.

F. Impact of Patch Size

The number of adjacent pixels used for HSI classification depends on the spatial size of HSI patches. This subsection studies the impact of spatial sizes on the performance of different methods. The spatial size of HSI patches is set from 3×3 to 13×13 to conduct experiments. OAs of 2-D CNN, DHCNet, SpecAN, SSAN and RIAN in testing samples at 180 rotation degrees are shown in Fig. 10. Since 1-D CNN and MGCN take hyperspectral pixels as the input, 1-D CNN and MGCN are not employed to conduct experiments. As the spatial size of HSI patches increases, the number of pixels that differ from the center pixel category in the HSI patch is also increasing. In the HSI patch, pixels of the same category as the central pixel can facilitate classification, but pixels of different categories from the center pixel can interfere with

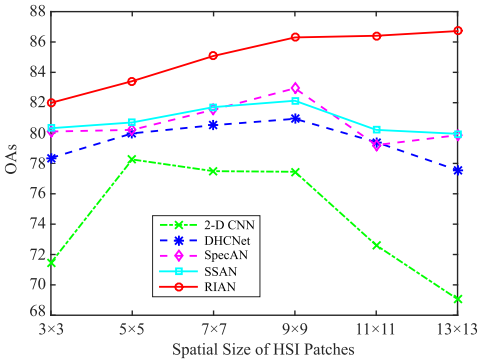

Fig. 10. OAs (%) of different methods with HSI patches of different spatial sizes on the Houston database.

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

classification [18], [48]. As shown in Fig. 10, as the patch size increases, the performance of RIAN increases. Experimental results demonstrate that RIAN can effectively suppress the interference from other categories of pixels.

V. Conclusion

In this paper, a rotation-invariant attention network is proposed for HSI classification. The proposed RIAN focuses on eliminating redundant spectral bands and exploring discriminative rotation-invariant spectral-spatial features of HSI. A center spectral attention module is proposed to suppress redundant spectral bands of input HSI patches. A rectified spatial attention module is proposed to extract rotation-invariant spectral-spatial features. Experiments demonstrate that RIAN outperforms recent spectral-spatial methods and is invariant to spatial rotation of HSI. However, different from the spectral-spatial convolution, the proposed rectified spatial attention module cannot effectively capture textural information of HSI, since the spatial aggregation is simply based on the similarity among pixels. In the future, we will study new spatial attention modules to explore textural information of HSI for classification.

Acknowledgment

The authors thank MindSpore for the partial support of this work, which is a new deep learning computing framework.[4]

References

[1] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[2] X. Du, X. Zheng, X. Lu, and A. A. Doudkin, "Multisource remote sensing data classification with graph fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10062–10072, Dec. 2021.

[3] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[4] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, early access, Apr. 13, 2022, doi: 10.1109/MGRS.2022.3145854.

[5] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[6] W. Chen, X. Zheng, and X. Lu, "Hyperspectral image super-resolution with self-supervised spectral–spatial residual network," *Remote Sens.*, vol. 13, no. 7, p. 1260, Mar. 2021.

[7] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[8] N. Audebert, B. L. Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[9] M. Imani and H. Ghassemian, "An overview on spectral and spatial information fusion for hyperspectral image classification: Current trends and challenges," *Inf. Fusion*, vol. 59, pp. 59–83, Oct. 2020.

[10] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[11] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[12] J. Peng *et al.*, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Mar. 2022.

[13] L. Fang, N. He, S. Li, P. Ghamisi, and J. A. Benediktsson, "Extinction profiles fusion for hyperspectral images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1803–1815, Mar. 2018.

[14] P. Ghamisi, M. D. Mura, and J. A. Benediktsson, "A survey on spectral–spatial classification techniques based on attribute profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2335–2353, May 2015.

[15] W. Sun, G. Yang, J. Peng, and Q. Du, "Lateral-slice sparse tensor robust principal component analysis for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 107–111, Jan. 2020.

[16] L. Mou and X. X. Zhu, "Learning to pay attention on spectral domain: A spectral attention module-based convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 110–122, Sep. 2020.

[17] X. Tang *et al.*, "Hyperspectral image classification based on 3-D octave convolution with spatial–spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

[18] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.

[19] F. Luo, L. Zhang, X. Zhou, T. Guo, Y. Cheng, and T. Yin, "Sparse-adaptive hypergraph discriminant analysis for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 6, pp. 1082–1086, Jun. 2020.

[20] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.

[21] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jul. 2015.

[22] H. Wu and S. Prasad, "Semi-supervised deep learning using pseudo labels for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1259–1270, Mar. 2018.

[23] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Apr. 2017.

[24] W. Li, F. Feng, H. Li, and Q. Du, "Discriminant analysis-based dimension reduction for hyperspectral image classification: A survey of the most recent advances and an experimental comparison of different techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 1, pp. 15–34, Mar. 2018.

[25] D. Hong *et al.*, "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[26] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[27] B. Fang, Y. Li, H. Zhang, and J. C.-W. Chan, "Hyperspectral images classification based on dense convolutional networks with spectral-wise attention mechanism," *Remote Sens.*, vol. 11, no. 2, p. 159, 2019.

[28] G. Cheng, J. Han, P. Zhou, and D. Xu, "Learning rotation-invariant and Fisher discriminative convolutional neural networks for object detection," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 265–278, Jan. 2019.

[29] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

[30] X. Zheng, T. Gong, X. Li, and X. Lu, "Generalized scene classification from small-scale datasets with multitask learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[31] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[32] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[33] H. Wu and S. Prasad, "Convolutional recurrent neural networks for hyperspectral data classification," *Remote Sens.*, vol. 9, no. 3, p. 298, 2017.

[34] B. Rasti *et al.*, "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[4]https://www.mindspore.cn/

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.

[35] D. Hong *et al.*, "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.

[36] J. Zhu, L. Fang, and P. Ghamisi, "Deformable convolutional neural networks for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 8, pp. 1254–1258, Aug. 2018.

[37] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[38] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[39] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[40] B. Deng, S. Jia, and D. Shi, "Deep metric learning-based feature embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1422–1435, Feb. 2020.

[41] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D-2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Jun. 2020.

[42] D. Hong *et al.*, "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[43] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[44] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[45] H. Zhao, J. Jia, and V. Koltun, "Exploring self-attention for image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10073–10082.

[46] N. Wang, Y. Zhang, and L. Zhang, "Dynamic selection network for image inpainting," *IEEE Trans. Image Process.*, vol. 30, pp. 1784–1798, 2021.

[47] X. Zheng, B. Wang, X. Du, and X. Lu, "Mutual attention inception network for remote sensing visual question answering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[48] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.

[49] H. Sun, S. Li, X. Zheng, and X. Lu, "Remote sensing scene classification by gated bidirectional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 82–96, Jan. 2020.

[50] X. Tang *et al.*, "Hyperspectral image classification based on 3-D octave convolution with spatial–spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

[51] J. Wang, J. Zhou, and W. Huang, "Attend in bands: Hyperspectral band weighting and selection for image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4712–4727, Dec. 2019.

[52] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Apr. 2020.

[53] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2019.

[54] Z. Xue, X. Yu, B. Liu, X. Tan, and X. Wei, "HResNetAM: Hierarchical residual network with attention mechanism for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3566–3580, 2021.

[55] K. Yang, H. Sun, C. Zou, and X. Lu, "Cross-attention spectral–spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[56] W. Ma, Q. Yang, Y. Wu, W. Zhao, and X. Zhang, "Double-branch multi-attention mechanism network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 11, p. 1307, Jun. 2019.

[57] H. Sun, X. Zheng, and X. Lu, "A supervised segmentation network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 2810–2825, 2021.

[58] W. Chen, X. Zheng, and X. Lu, "Semisupervised spectral degradation constrained network for spectral super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[59] X. Zheng, W. Chen, and X. Lu, "Spectral super-resolution of multispectral images using spatial–spectral residual attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[60] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Jun. 2018, pp. 7794–7803.

[61] H. Hu, Z. Zhang, Z. Xie, and S. Lin, "Local relation networks for image recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3463–3472.

[62] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[63] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

Xiangtao Zheng (Member, IEEE) received the M.Sc. and Ph.D. degrees in signal and information processing from the University of Chinese Academy of Sciences, China, in 2014 and 2017, respectively. He is currently an Associate Professor with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shaanxi, China. His main research interests include computer vision and pattern recognition.

Hao Sun graduated from the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shaanxi, China, and the University of Chinese Academy of Sciences, Beijing, China. He received the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences in 2021. His current research interests include pattern recognition, computer vision, and machine learning.

Xiaoqiang Lu (Senior Member, IEEE) is currently a Full Professor with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shaanxi, China. His current research interests include pattern recognition, machine learning, hyperspectral image analysis, cellular automata, and medical imaging.

Wei Xie is currently a Professor with the School of Computer Science, Central China Normal University, Wuhan, Hubei, China. His research interests include image processing, computer vision, and deep learning.

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on May 25,2023 at 01:11:28 UTC from IEEE Xplore. Restrictions apply.